

Geometry

Unit 3

Lesson 1 Practice

Name **Answer Key** _____

Date _____

Period _____

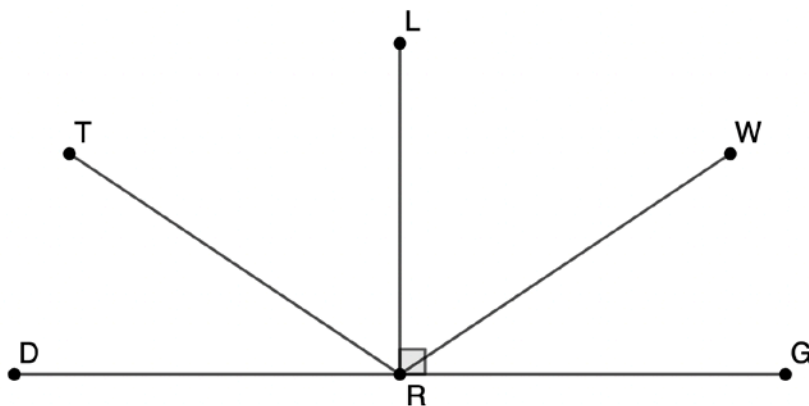
Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank

Definition of congruence	Linear Pair Postulate
Supplements of congruent angles are congruent.	Complements of congruent angles are congruent.
Angle addition postulate	Substitution property of equality
Definition of supplementary angles	Definition of complementary angles

Given: $\angle DRT \cong \angle GRW$, $m\angle DRL = 90^\circ$, $m\angle GRL = 90^\circ$

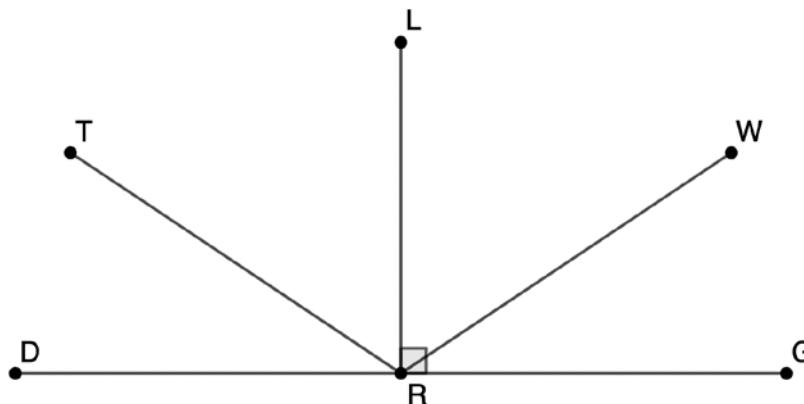
Prove: $\angle LRT \cong \angle LRW$



Complete the table using the word bank to prove $\angle LRT \cong \angle LRW$.

Statement	Reason
$m\angle DRL = 90^\circ$	Given
$m\angle LRG = 90^\circ$	Given
$m\angle DRT + m\angle LRT = m\angle DRL$	Angle addition postulate
$m\angle DRT + m\angle LRT = 90^\circ$	1. Substitution property of equality
$\angle DRT$ and $\angle LRT$ are complementary angles	2. Definition of complementary angles
$m\angle LRW + m\angle GRW = m\angle LRG$	Angle addition postulate
$m\angle LRW + m\angle GRW = 90^\circ$	3. Substitution property of equality
$\angle LRW$ and $\angle GRW$ are complementary angles	4. Definition of complementary angles
$\angle DRT \cong \angle GRW$	Given
$\angle LRT \cong \angle LRW$	5. Complements of congruent angles are congruent

Lines $\overline{DG}$, $\overline{TR}$, $\overline{LR}$, and $\overline{RW}$ are shown where $\angle DRT \cong \angle GRW$, $m\angle DRL = 90^\circ$, $m\angle GRL = 90^\circ$ and $m\angle DRT = \left(\frac{5}{2}w - 8\right)^\circ$ and $m\angle GRW = (2w - 1)^\circ$.



6. Find the value of w .

$$w = 14$$

7. What is $m\angle DRT$?

$$m\angle DRT = 27^\circ$$

8. What is $m\angle DRW$?

$$m\angle DRW = 153^\circ$$

9. What is $m\angle LRW$?

$$m\angle LRW = 63^\circ$$

10. What is $m\angle TRG$?

$$m\angle TRG = 153^\circ$$

Geometry

Unit 3

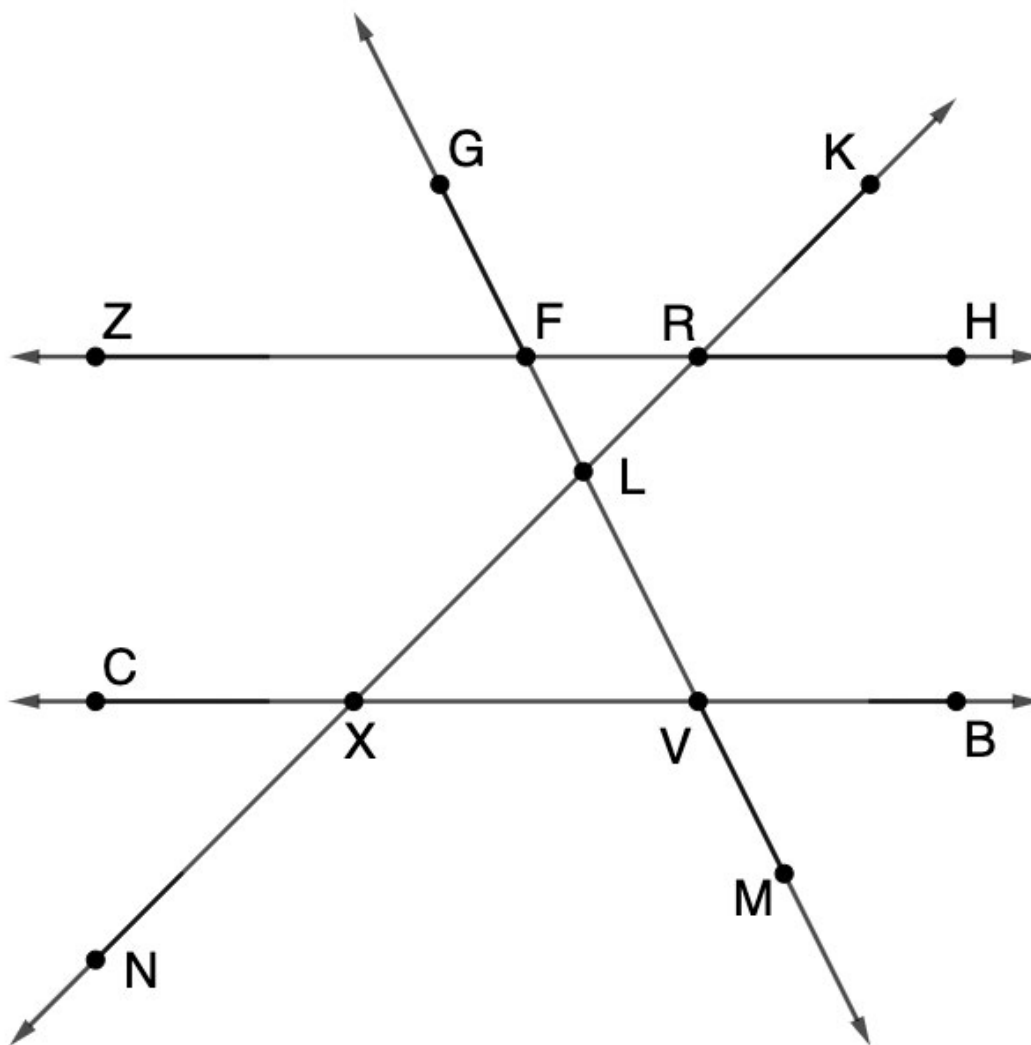
Lesson 2 Practice

Name **Answer Key** _____

Date _____

Period _____

$\overleftrightarrow{ZH}$, $\overleftrightarrow{CB}$, $\overleftrightarrow{NK}$ and $\overleftrightarrow{GM}$ are shown where $\overleftrightarrow{ZH} \parallel \overleftrightarrow{CB}$, $m\angle ZRL = 40^\circ$, $m\angle GLR = 64^\circ$, $m\angle GFZ = (3x + 10)^\circ$, and $m\angle ZFV = (5x - 6)^\circ$.



1. What is the value of x ?

$$x = 22$$

Determine the measure of the angles.

2. $m\angle GFZ$ 76°

7. $m\angle CXN$ 40°

3. $m\angle ZFV$ 104°

8. $m\angle XVB$ 180°

4. $m\angle LFR$ 76°

9. $m\angle LXB$ 40°

5. $m\angle RLV$ 116°

10. $m\angle MVB$ 76°

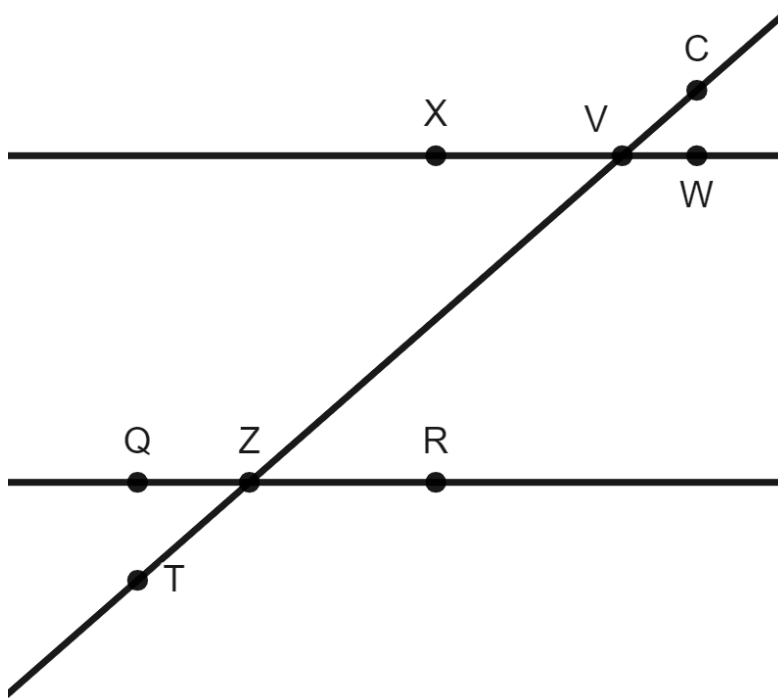
6. $m\angle XLV$ 64°

Geometry
Unit 3
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

Lines $\overline{XW}$, $\overline{QR}$, and $\overline{TC}$ are shown.

$\overline{XW} \parallel \overline{QR}$, $m\angle CVW = 27^\circ$



Complete the table.

Angle	Angle Measure	Relationship with $\angle CVW$
$\angle XVZ$	1. 27°	2. Vertical angles
$\angle VZR$	3. 27°	4. Corresponding angles
$\angle QZT$	5. 27°	6. Alternate exterior angles

$\angle CVW$ and $\angle CVX$ are supplementary, so $m\angle CVX = 153^\circ$.

Complete the table.

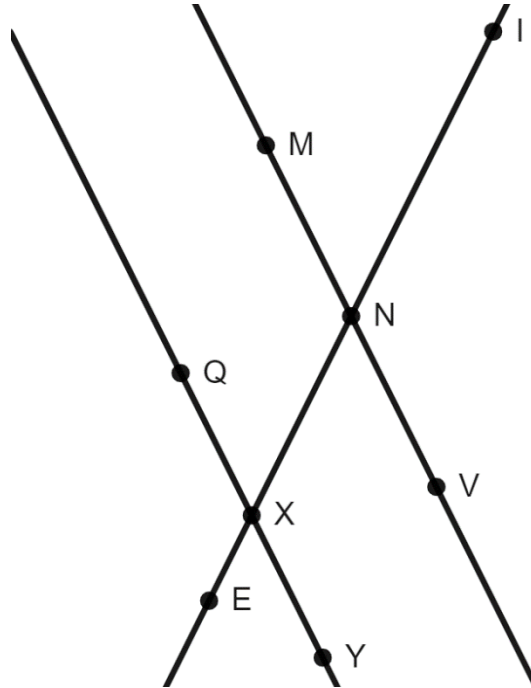
Angle	Angle Measure	Relationship with $\angle CVX$
$\angle ZVW$	7. 153°	8. Vertical angles
$\angle TZR$	9. 153°	10. Alternate exterior angles

Geometry
Unit 3
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Lines $\overline{MV}$, $\overline{QY}$, and $\overline{EI}$ are shown.

$\overline{MV} \parallel \overline{QY}$, $m\angle MNX = 130^\circ$



Identify the relationship between each pair of angles.

Angles	Congruent or Supplementary?
$\angle MNX$ and $\angle QXN$	1. Supplementary
$\angle MNX$ and $\angle MNI$	2. Supplementary
$\angle MNX$ and $\angle VNI$	3. Congruent
$\angle MNX$ and $\angle QXE$	4. Congruent

Complete the table according to the diagram.

Angle	Angle Measure	Relationship with $\angle EXY$
$\angle VNI$	5. 130°	6. Consecutive exterior angles
$\angle YXN$	7. 130°	8. Linear pair

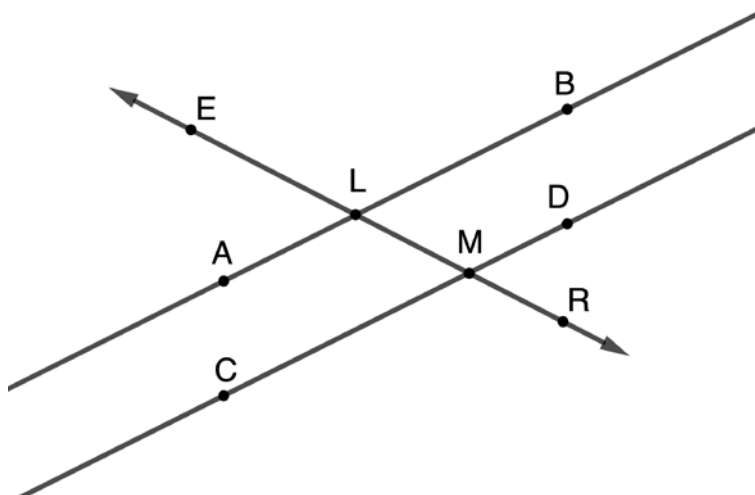
9. If $m\angle MNX = (10a)^\circ$ and $m\angle QXN = (4a - 2)^\circ$, what is the value of a ?
 $a = 13$

10. If $m\angle EXY = (4w - 6)^\circ$ and $m\angle VNI = (9w + 4)^\circ$, what is the value of w ?
 $w = 14$

Geometry
Unit 3
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

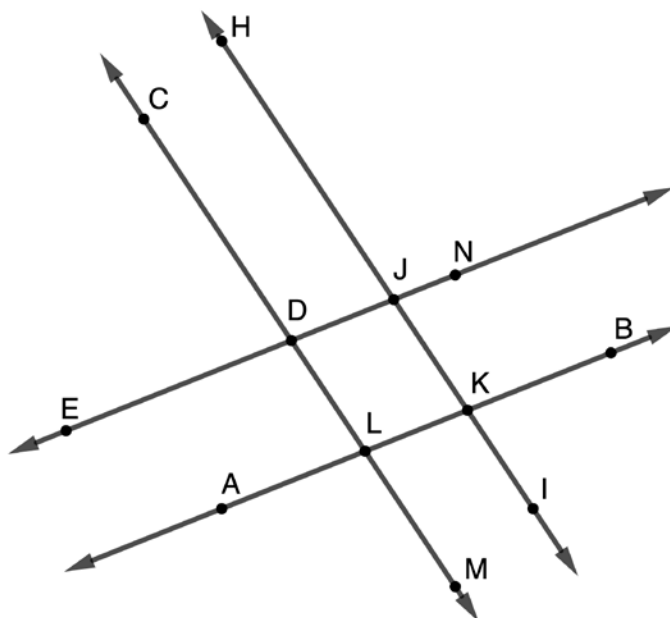
1. Three lines are shown, $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, and $\overleftrightarrow{ER}$. $\overleftrightarrow{ER}$ intersects $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$.



Determine if the given condition could be used to justify that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$?	Justification
If $m\angle ELA = 78^\circ$, then $m\angle DMR = 78^\circ$	☒ Yes ☐ No	If two lines are intersected by a transversal so the alternate exterior angles are congruent, then the lines are parallel.
$\angle ALM \cong \angle CMR$	☒ Yes ☐ No	If two lines are intersected by a transversal so the corresponding angles are congruent, then the lines are parallel.
$m\angle ELA + m\angle ELB = 180^\circ$	☐ Yes ☒ No	Two supplementary adjacent angles do not form a set of two parallel lines.
$\angle ELB \cong \angle CMR$	☒ Yes ☐ No	If two lines are intersected by a transversal so the alternate exterior angles are congruent, then the lines are parallel.
$\angle LMC \cong \angle DMR$	☐ Yes ☒ No	Congruent vertical angles do not prove parallel lines
If $m\angle ELB = 95^\circ$, then $m\angle ELA = 85^\circ$	☐ Yes ☒ No	Two supplementary adjacent angles do not form a set of two parallel lines.
If $m\angle DMR = 81^\circ$, then $m\angle ALM = 99^\circ$	☒ Yes ☐ No	Since $m\angle DMR + m\angle CMR = 180^\circ$ because they form a linear pair, then $m\angle CMR = 99^\circ$. Since $m\angle ALM = 99^\circ$ and $m\angle CMR = 99^\circ$, $m\angle ALM = m\angle CMR$ by transitive property, and $\angle ALM \cong \angle CMR$ by the definition of congruence and since the corresponding angles are congruent, the lines intersected by the transversal must be parallel.

2. $\overleftrightarrow{CM}$, $\overleftrightarrow{HI}$, $\overleftrightarrow{EN}$, and $\overleftrightarrow{AB}$ are shown, where $m\angle CDE = 90^\circ$.



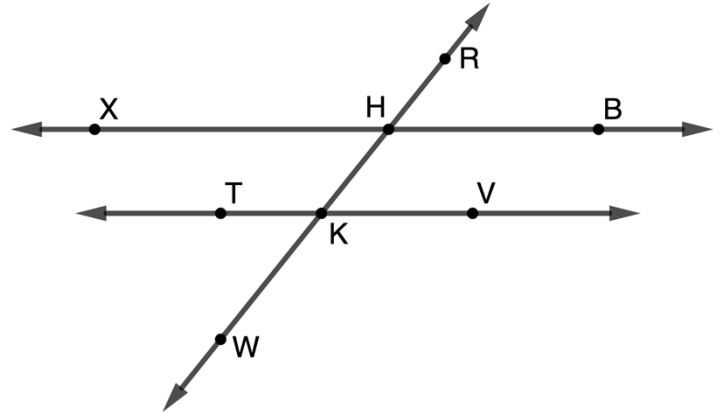
Determine if the given conditions could be used to justify that $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$.

Condition	Is $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$?	Justification
$\angle CDE \cong \angle HJN$	☐ Yes ☒ No	If two lines are intersected by a transversal so the consecutive exterior angles are congruent, that is not enough information to justify congruence.
$\angle CDJ \cong \angle DLK$	☒ Yes ☐ No	If two lines are intersected by a transversal so the corresponding angles are congruent, then the lines are parallel.
$m\angle EDL + m\angle NJK = 180^\circ$	☒ Yes ☐ No	If two lines are intersected by a transversal so the consecutive exterior angles are supplementary, then the lines are parallel.
$\angle KJD \cong \angle HJN$	☐ Yes ☒ No	Congruent vertical angles do not prove that two lines are parallel.
$\angle JKL \cong \angle BKI$	☐ Yes ☒ No	Congruent vertical angles do not prove that two lines are parallel.
$m\angle BKI + m\angle LKI = 180^\circ$	☐ Yes ☒ No	Line pair of angles do not prove that two lines are parallel.

Geometry
Unit 3
Lesson 6 Practice

Name **Answer Key** _____
Date _____
Period _____

$\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ is shown where $m\angle RHB = (5x + 1)^\circ$
and $m\angle KHX = (6x - 10)^\circ$.



1. Determine the value of x .

$x = 11$

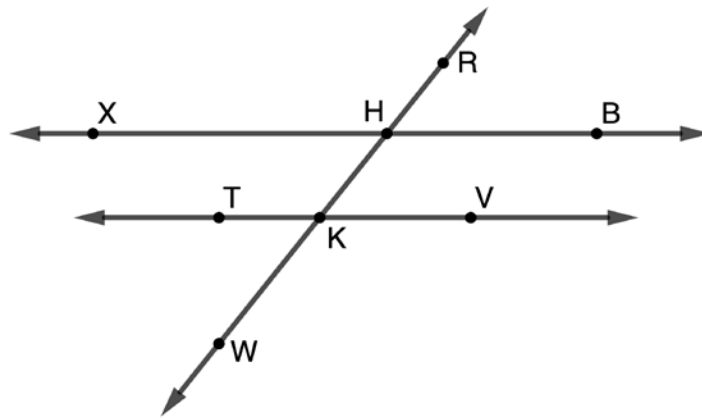
2. What is $m\angle RHB$?

56°

3. What is $m\angle VKW$?

124°

$\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ is shown where $m\angle XHK = (8x + 6)^\circ$ and $m\angle VKW = 126^\circ$.



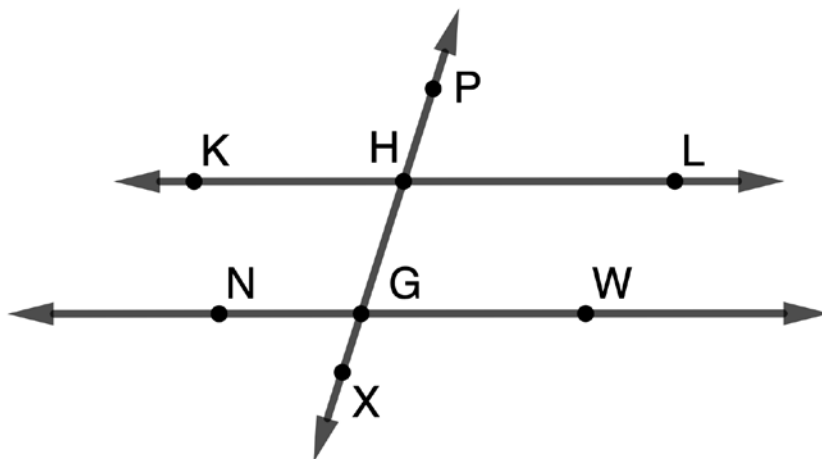
4. Determine the value of x .

$x = 6$

5. What is $m\angle HKV$?

54°

$\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$ is shown where $m\angle KHG = (4s + 6)^\circ$, $m\angle HGW = (6s - 16)^\circ$, and $m\angle NGX = (y + 9)^\circ$.



6. Determine the value of s .

$$s = 11$$

7. Determine the value of y .

$$y = 41$$

8. What is $m\angle KHG$?

$$50^\circ$$

9. What is $m\angle HGW$?

$$50^\circ$$

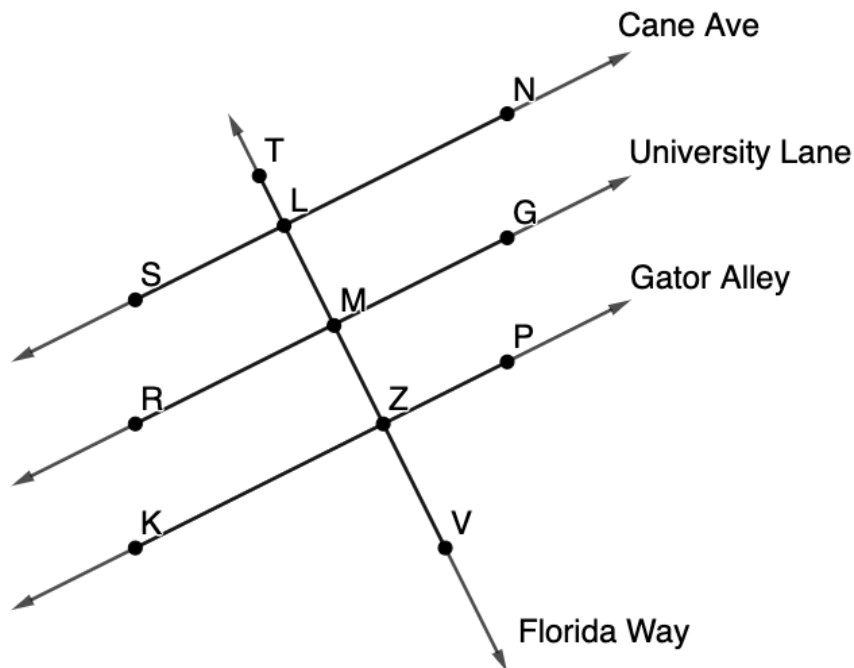
10. What is $m\angle NGH$?

$$130^\circ$$

Geometry
Unit 3
Lesson 7 Practice

Name **Answer Key** _____
Date _____
Period _____

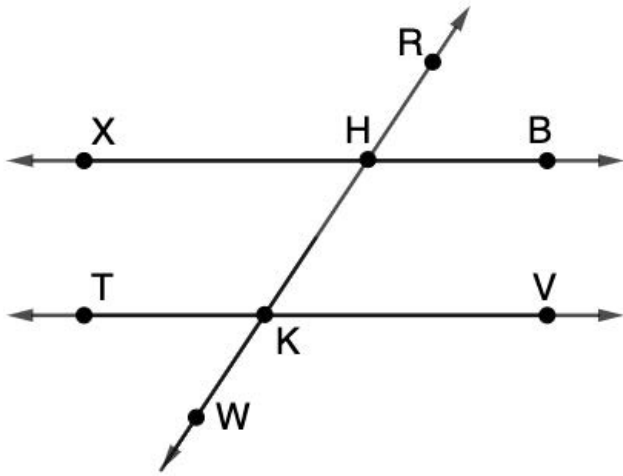
Streets Cane Ave, $\overleftrightarrow{SN}$, University Lane, $\overleftrightarrow{RG}$, and Gator Alley, $\overleftrightarrow{KP}$, are parallel to each other where Florida Way, $\overleftrightarrow{TV}$, crosses each of them as shown.



Determine the value of the variables and each angle measure.

- $m\angle LMG = (9x + 18)^\circ$ and $m\angle PZV = (2x + 19)^\circ$.
 $9x + 18 + 2x + 19 = 180$
 $11x + 37 = 180$
 $x = 13$
 $m\angle LMG = 9(13) + 18 = 135^\circ$
 $m\angle PZV = 2(13) + 19 = 45^\circ$
- $m\angle TLS = (9x + 19)^\circ$, $m\angle TLN = (5x - 7)^\circ$, and $m\angle NLM = (y - 1)^\circ$.
 $9x + 19 + 5x - 7 = 180$
 $14x + 12 = 180$
 $x = 12$
 $m\angle TLS = 9(12) + 19 = 127^\circ$
 $m\angle TLN = 5(12) - 7 = 53^\circ$
 $m\angle NLM = 127^\circ = y - 1$
 $128 = y$
- $m\angle LMR = (9x - 7)^\circ$, $m\angle LMG = (y + 16)^\circ$, and $m\angle GMZ = (7x + 19)^\circ$.
 $9x - 7 = 7x + 19$
 $2x = 26$
 $x = 13$
 $m\angle LMR = 9(13) - 7 = 110^\circ$
 $m\angle GMZ = 7(13) + 19 = 110^\circ$
 $m\angle LMR + m\angle LMG = 180^\circ$
 $110 + y + 16 = 180$
 $y = 54$
 $m\angle LMG = 54 + 16 = 70^\circ$

Lines $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$ and $\overleftrightarrow{RW}$ are shown where $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$.



Determine the value of the variables and each angle measure.

4. $m\angle TKW = (2y - 5)^\circ$, $m\angle HKV = (9x - 19)^\circ$, and $m\angle RHB = (8x - 11)^\circ$.

$$9x - 19 = 8x - 11$$

$$x = 8$$

$$2y - 5 = 53$$

$$y = 29$$

$$m\angle HKV = 9(8) - 19 = 53^\circ$$

$$m\angle RHB = 8(8) - 11 = 53^\circ$$

$$m\angle TKW = 53^\circ$$

5. $m\angle RHB = (4x + 19)^\circ$ and $m\angle XHK = (8x + 7)^\circ$.

$$4x + 19 = 8x + 7$$

$$x = 3$$

$$m\angle RHB = 4(3) + 19 = 31^\circ$$

$$m\angle XHK = 8(3) + 7 = 31^\circ$$

6. $m\angle HKV = (x - 5)^\circ$ and $m\angle BHK = (2x + 14)^\circ$.

$$x - 5 + 2x + 14 = 180$$

$$3x + 9 = 180$$

$$x = 57$$

$$m\angle HKV = 57 - 5 = 52^\circ$$

$$m\angle BHK = 2(57) + 14 = 128^\circ$$

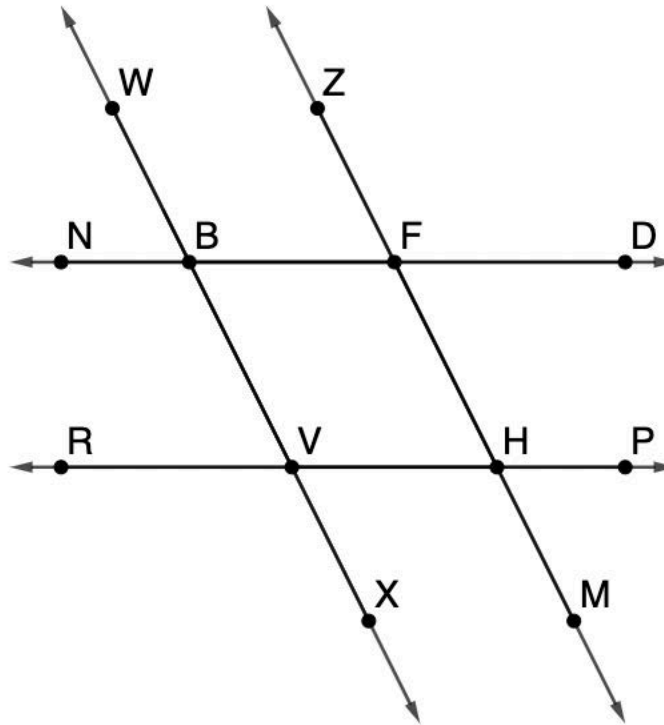
7. $m\angle KHX = (6x - 5)^\circ$ and $m\angle WKT = 101^\circ$.

$$6x - 5 = 101$$

$$x = 17\frac{2}{3}$$

$$m\angle KHX = 6(17\frac{2}{3}) - 5 = 101^\circ$$

Lines $\overleftrightarrow{ND}$, $\overleftrightarrow{RP}$, $\overleftrightarrow{WX}$ and $\overleftrightarrow{ZM}$ are shown.



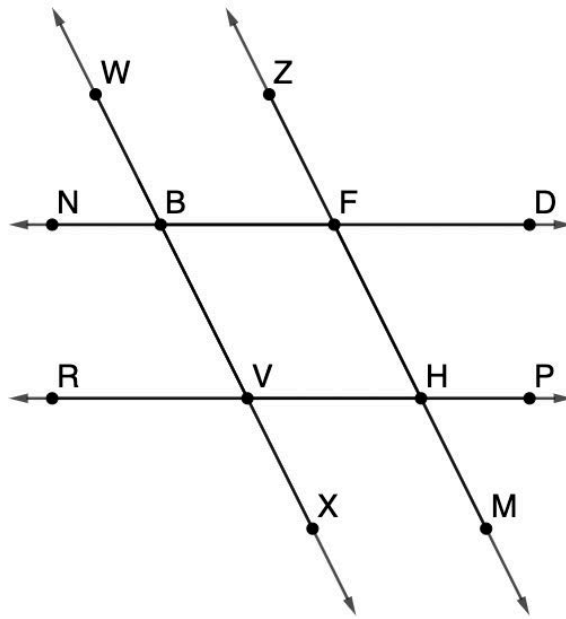
Circle the contradictory piece of information in each row.

	Item 1	Item 2	Item 3	Item 4
1.	$\angle NBW \cong \angle MHP$	$\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$	$\overleftrightarrow{WX} \nparallel \overleftrightarrow{ZM}$	$\angle BFZ \cong \angle HFD$
2.	$\angle ZFD$ and $\angle NBV$ are supplementary	$m\angle ZFD = 118^\circ$	$\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$	$\angle RVX \cong \angle FHP$
3.	$\angle BFH \cong \angle HVB$	$\angle NBW$ and $\angle MHP$ are supplementary	$\angle NBV \cong \angle ZFD$	$\angle ZFD$ and $\angle MHP$ are supplementary
4.	$\angle WBN \cong \angle VBF$	$\angle RVX$ and $\angle NBW$ are supplementary	$\angle BFZ \cong \angle VHM$	$\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$
5.	$\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$	$m\angle WBF = 203^\circ$	$\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$	$\angle ZFD$ and $\angle RVB$ are supplementary

Choose from the bank of statements and reasons to complete the paragraph proof.

Given: $\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$ and
 $\angle ZFD \cong \angle VHM$

Prove: $\angle DFH \cong \angle BVR$



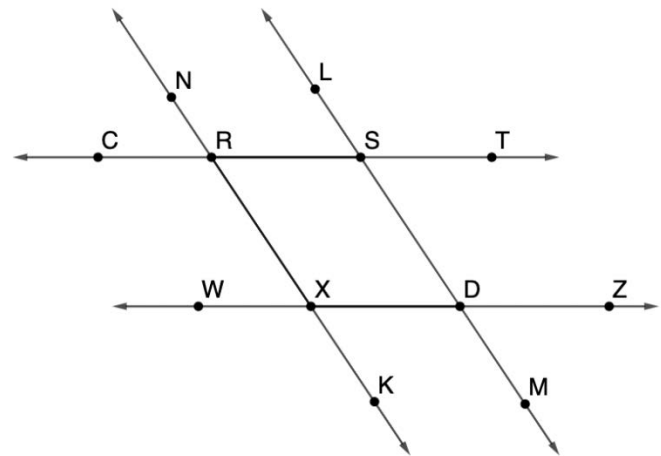
It is given that 6. $\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$. Therefore, $\angle BVR \cong \angle FHV$ because if two parallel lines are intersected by a transversal then 7. corresponding angles are congruent. It is also given that 8. $\angle ZFD \cong \angle VHM$. If 9. alternate interior angles are congruent, then the two lines that are intersected by a transversal are parallel and it can be written that $\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$. Then, by the alternate interior angles theorem, it must be true that 10. $\angle DFH \cong \angle FHV$. Therefore, by transitive property of congruence, $\angle DFH \cong \angle BVR$.

Statements and Reasons	
$\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$	corresponding angles are congruent
$\angle DFH \cong \angle BVR$	a linear pair is supplementary
$\angle DFH \cong \angle FHV$	vertical angles are congruent
$\overleftrightarrow{ND} \parallel \overleftrightarrow{WX}$	alternate interior angles are congruent
$\angle WBF$ and $\angle NBV$	alternate exterior angles are congruent
$\angle ZFD \cong \angle VHM$	transitive property of congruence

Complete the two-column proof.

Given: $\overleftrightarrow{NK} \parallel \overleftrightarrow{LM}$ and $\angle LST \cong \angle WXK$

Prove: $\angle CRX \cong \angle RXD$



Statement	Reason
1. $\overleftrightarrow{NK} \parallel \overleftrightarrow{LM}$	1. Given
2. $\angle WXK \cong \angle XDM$	2. If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
3. $\angle LST \cong \angle WXK$	3. Given
4. $\angle LST \cong \angle XDM$	4. Transitive Property of Congruence
5. $\overleftrightarrow{CT} \parallel \overleftrightarrow{WZ}$	5. If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.
6. $\angle CRX \cong \angle RXD$	6. If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

Write the statements and reasons in the correct order in the two-column proof.

$\angle CRX \cong \angle RXD$ Given

$\angle WXK \cong \angle XDM$ Transitive Property of Congruence

$\angle LST \cong \angle XDM$ If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

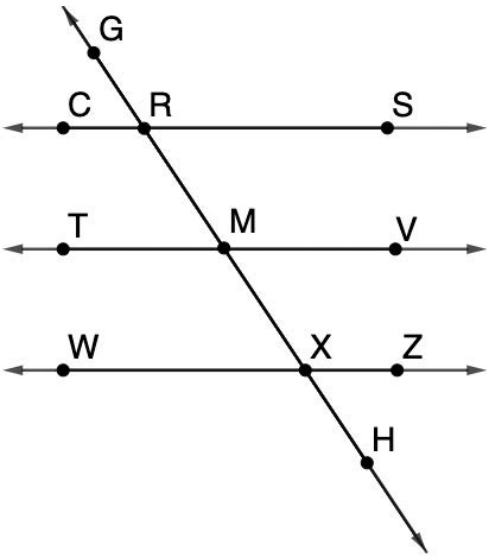
$\angle LST \cong \angle WXK$ If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.

$\overleftrightarrow{CT} \parallel \overleftrightarrow{WZ}$ If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.

Complete the two-column proof.

Given: $\angle RMV \cong \angle MXZ$ and
 $\angle GRC$ and $\angle WXH$ are supplementary

Prove: $\angle TMX \cong \angle GRS$



Statement	Reason
1. $\angle RMV \cong \angle MXZ$	1. Given
2. $\overleftrightarrow{TV} \parallel \overleftrightarrow{WZ}$	2. If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.
3. $\angle GRC$ and $\angle WXH$ are supplementary	3. Given
4. $\angle MXW$ and $\angle WXH$ are supplementary	4. If two angles form a linear pair, then they are supplementary.
5. $\angle GRC \cong \angle MXW$	5. Supplements of congruent angles are congruent.
6. $\overleftrightarrow{CS} \parallel \overleftrightarrow{WZ}$	6. If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
7. $\angle TMX \cong \angle GRS$	7. If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent

Write the statements and reasons in the correct order in the two-column proof.

$\overleftrightarrow{TV} \parallel \overleftrightarrow{WZ}$	Supplements of congruent angles are congruent.
$\overleftrightarrow{CS} \parallel \overleftrightarrow{WZ}$	If two angles form a linear pair, then they are supplementary.
$\angle TMX \cong \angle GRS$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle GRC \cong \angle MXW$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\angle MXW$ and $\angle WXH$ are supplementary	If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.